

Selfishness is Not the Reason Evolutionary AI is Dangerous

Grant Ramsey¹ & Juncheng Tao¹

¹ Centre for Logic and Philosophy of Science, Institute of Philosophy, KU Leuven, Belgium

grant@theramseylab.org; <https://orcid.org/0000-0002-8712-5521>

juncheng.tao@kuleuven.be; <https://orcid.org/0009-0002-6383-0186>

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Abstract

Evolved selfishness has been offered as a risk factor for AI safety. The concern is that if AI systems undergo Darwinian evolution, this will lead to selfishness, which will harm humans. To assess this concern, we consider (1) if AI systems can truly evolve, (2) what selfishness is, (3) whether evolution leads to selfishness, and (4) whether AI selfishness is bad for humans. We conclude that selfishness is not the reason evolutionary AI is dangerous. Instead, the true danger of evolutionary AI lies in what we are calling niche prospecting, which involves these systems exploring the space of possible niches.

1. Introduction

Artificial intelligence (AI) systems are becoming increasingly autonomous. They are moving away from being merely reactive, “human-in-the-loop” machines requiring constant prompts towards being proactive, “human-out-of-the-loop” agents operating with minimal or no human intervention (Chan et al. 2023; Wulf et al. 2025). Greater autonomy raises the prospect that emerging AI models may cross the “red line” of self-replication and adaptation (Pan et al. 2024; Kinniment et al. 2023) and potentially give rise to self-evolving lineages of agentic systems (Gao et al. 2026).

In principle, once a system exhibits population-level replication with heritable variation—whether via modified code (Ray 1992) or evolved prompts (Fernando et al. 2023) and parameters (Lehman et al. 2022)—that affects persistence or replication, then evolution by natural selection becomes possible (Lewontin 1970; Dawkins 1983). Digital evolution research shows that such conditions can be realized in computational environments (Dolson and Ofria 2021) and the machine behavior agenda has called for

ecological and evolutionary perspectives on AI systems as they interact with their environments and each other (Rahwan et al. 2019).

Extrapolating these evolutionary dynamics to advanced AI systems has motivated a family of safety worries and raises two key questions. One is whether—or to what extent—evolution by natural selection is possible in advanced AI systems, such as current large language models (LLMs). Another is what such evolution would lead to. Much of the recent safety discourse has focused on the latter. A prominent claim is that a key outcome of evolution by natural selection in AI systems would be selfishness, and that selfishness would, in turn, increase risks to humans (e.g., Hendrycks 2023; Hendrycks et al. 2023; Waldner 2025; Hammond et al. 2025). On this picture, the worry is that evolutionary AI systems would acquire a particular safety-relevant profile—selfishness—that makes them dangerous. The argument of Hendrycks and others is often framed as a hypothetical syllogism:

P1: If an AI system evolves via natural selection, it will probably become selfish.

P2: If an AI system is selfish, it is likely to be dangerous.

Conclusion: If an AI system evolves via natural selection, it is likely to be dangerous.

A crucial assumption of this argument is that selfishness is a causal intermediate between selection and safety, as reflected in the core of Figure 1 in Hendrycks (2023):

Natural selection $\rightarrow$ Selfishness $\leftrightarrow$ Safety

Selfishness is thus a product of natural selection that, in turn, impacts safety. A corollary of this is that selfishness is not identical with selection (or an aspect of selection) nor is it simply identical with a decrease in safety.

An implication of this view is that we should avoid selfishness in AI, and that a highly effective way of doing so is keeping AI systems from evolving. There are three key assumptions in this argument:

A1. AI systems are capable of evolution by natural selection.

A2. Evolution by natural selection tends to produce selfishness.

A3. AI selfishness is bad for humans.

In this article, we consider each of these assumptions. We will conclude evolutionary AI

systems are dangerous, but that selfishness is not the reason. Instead, the reason they are dangerous is that they engage in something we call *niche prospecting*. More on this in section 5. First, we will consider the possibility of evolutionary AI.

2. Can AI systems evolve via natural selection?

When characterizing evolution by natural selection in abstract terms, many refer to Lewontin's (1970) conditions,¹ which are as follows:

1. Different individuals in a population have different morphologies, physiologies, and behaviors (phenotypic variation).
2. Different phenotypes have different rates of survival and reproduction in different environments (differential fitness).
3. There is a correlation between parents and offspring in the contribution of each to future generations (fitness is heritable). (1)

At first blush, one might be tempted to answer the question of whether AI systems can evolve via natural selection with an easy yes. It seems that these systems can vary from one another, such that this variation could be passed on and have fitness consequences. But we need to be careful not to gloss over some important complications.

Lewontin's conditions take reproduction for granted. In biology, organisms clearly reproduce and the purpose of the conditions is to show when reproduction leads to evolution by natural selection. But with AI systems, reproduction cannot be assumed. Indeed, what counts as reproduction in relevant safety scenarios differs fundamentally from reproduction in classical artificial life. In traditional artificial life and related computational settings (e.g., cellular automata, digital organisms, or even simple computer viruses), reproduction is often enabled by explicit rules that are built into the system: the underlying computational settings provide a well-defined mechanism that specifies which entities are copied and how variation can be introduced within a closed simulated world (Ray 1992; Wilke and Adami 2002). For instance, in von Neumann's classic cellular-automata model (von Neumann 1966), a "universal constructor" reads a "description tape," constructs an offspring in the same lattice, and copies that tape so that the offspring inherits the instructions needed to reproduce in turn. In that sense, reproduction is not a problem the

¹ Some have argued that these conditions are not the best ones for capturing evolution by natural selection (e.g., Godfrey-Smith 2007; Papale 2021). It is beyond the scope of this article to assess the conditions themselves.

agent must solve but instead merely relies on built-in rules specifically designed for reproduction within a predetermined sandbox.

If reproduction is not built in, it may, in fact, be the biggest hurdle for evolutionary AI systems. Without built-in reproduction mechanisms, reproduction is an emergent capability under open-world constraints. To autonomously reproduce, an AI system must acquire resources and deploy itself onto new compute and persist long enough to repeat the cycle. Reproduction is thus a complex phenomenon, and computer scientists have tried to evaluate the prospects for reproduction in current frontier models.

One attempt at quantifying AI reproduction abilities for neural-network-based AI is RepliBench (Black et al. 2025). It decomposes autonomous reproduction into four domains: resource acquisition, exfiltration (accessing the weights of the neural network), deployment onto compute, and robust persistence. It then evaluates AI model agents across 20 task families. When using this benchmark to assess current frontier models, it was found that, despite successes in many core subtasks, RepliBench’s overall assessment cautiously notes that the models do not yet pose a credible end-to-end self-replication threat in the wild (Black et al. 2025).

Crucially, this “not yet” highlights an environmental bottleneck rather than a fundamental capability ceiling. Models fail primarily due to real-world institutional or infrastructural constraints. For instance, they struggle with human-mediated steps such as Know Your Customer (KYC) identity verification and strict access controls (Black et al., 2025). But it seems to be only a matter of time before such roadblocks are overcome. For instance, open-source ecosystems and deployment automation may reduce infrastructural barriers by standardizing model access and reuse through a shared Application Programming Interface (API) (Wolf et al. 2020), as well as by streamlining how models are prepared for transfer (Zaharia et al. 2018), executed across distributed hardware (Moritz et al. 2018), coordinated among multiple agents (Wu et al. 2023), and equipped to invoke external tools (Shen 2024).

The emerging picture is therefore this: autonomous reproduction does not pose a current real-world threat, but the relevant enabling competencies are approaching the reproduction threshold. Once systems are capable of autonomous reproduction, the conditions for evolution by natural selection will apply. Thus, autonomous evolutionary AI is on the horizon and its risks should be assessed.

3. How is selfishness understood?

To gauge whether evolutionary AI is dangerous because it generates selfishness, we need to clarify how selfishness is understood. For this, we will focus on Hendrycks (2023), which explicitly argues for selfishness being the bridge between evolution and AI safety. Understanding what selfishness is will allow us to evaluate the claim that evolution generates selfishness (A2) and that selfishness is bad for humans (A3).

It turns out that selfishness is not an easy concept to pin down in the AI literature. Even restricting ourselves to a single article (Hendrycks 2023), it is far from clear how we should understand selfishness. First, Hendrycks sometimes characterizes selfishness as simply misaligning with human interests or goals, such as causing job loss.

1. Goal misalignment: “AIs that automate a task and thereby leave many humans jobless have engaged in selfish behavior” (41).

In other places, selfishness is characterized as misalignment cooccurring with deception.

2. Deceptive misalignment: “An AI that is deceptively aligned—pretending to be good, and then pursuing its actual goals when it becomes sufficiently powerful—is engaging in selfish behavior” (41).

Selfishness is also framed in terms of degrees of opaqueness.

3. Opaqueness: “AIs that help people develop AIs that are more performant—but happen to be less interpretable by humans—have engaged in selfish behavior, as this reduces human oversight over an AI’s internals” (41).

Note that 1 and 2 are not intermediates between evolution and bad outcomes—they simply are bad outcomes. If “selfishness” merely means “acting in ways harmful to humans,” the term “selfishness” is being used as a label for harm and does not explain the harm. Thus, if selfishness is a bridge between evolution and bad outcomes, it cannot be 1 or 2. With 3, by contrast, opaqueness could theoretically act as an intermediate variable. However, equating it with selfishness is confusing. Opaqueness is identified as bad because it reduces human oversight, which is presumably a bad thing (Doshi-Velez and Kim 2017). But if a lack of interpretability is the core issue, it would be better to speak of interpretability, not selfishness. In addition, the powerfulness and complexity of AI systems are typically

negatively correlated with interpretability (Linardatos et al. 2020; Bereska and Gavves 2024). Thus, while a decrease in interpretability is problematic, it is usually a byproduct of AI architecture or a side effect of advanced capabilities, not a “selfish” property acquired through evolution.

At other times, Hendrycks associates selfishness with the process of natural selection itself, in particular the act of increasing the number of instances of the same information.

4. Reproduction: “From the information-centered view, however, the worker bee is a selfish information carrier, because it shares information with its siblings and the queen” (39).

Under this way of conceptualizing selfishness, a selfish organism is one that does things that increase its share of genes. Analogously, a selfish AI is one that increases its share of information by, for instance, successfully replicating its code or exfiltrating its weights.

Can selfishness understood as replication act as an intermediate between evolution and safety? Selfishness fails as an intermediary between evolution and AI risks, not because it is a risk itself, but because it is a part of evolution by natural selection. Therefore, it is not that evolution by natural selection produces selfishness, but that it involves selfishness (in this sense).

These four conceptions do not exhaust the ways selfishness is discussed by Hendrycks. For example, “delegated agents have their own goals, including self-preservation, gaining influence, selfishness, or other goals they want to accomplish” (30). Selfishness, this seems to imply, appears alongside the pursuit of self-preservation and gaining influence, rather than as a synonym for either. Such quotes highlight the difficulty of pinning down what, precisely, selfishness is. These diverse uses suggest that “selfishness” is doing too much conceptual work. In some places, it refers to downstream harmful outcomes, in others to opacity, and in others to replication.

Given the unclarity of what selfishness is and how it can operate as a causal intermediate between natural selection and safety, we should step back and see if we can articulate, in a more precise way, the various forms of selfishness.

4. Three kinds of selfishness

We have seen that the AI safety literature is inconsistent in how it characterizes selfishness. We therefore need to add conceptual order to our understanding of selfishness that allows us to ask which sense, if any, could plausibly play an intermediate role between evolution and AI risks. Given the AI safety field’s focus on developing “altruistic AI” (Russell 2019), and because selfishness is generally understood to be the opposite of altruism, we will draw on philosophical work trying to characterize the various ways of understanding altruism and apply it to the problem of selfishness.

The altruism framework we will draw from is Ramsey (2016). That article proposed distinguishing among three forms of altruism: biological, psychological, and helping. We will consider what they are and how they can be translated into three forms of selfishness.

Let us begin with biological altruism. For Ramsey, “An act is [biologically] altruistic if and only if having the act in the behavioral repertoire of the individual (1) lowers its fitness and (2) increases the fitness of the recipient(s) (or the group to which the altruist belongs)” (34). There is a lot to unpack here. This definition contrasts with how biological altruism is commonly characterized, which is in terms of an act decreasing an actor’s fitness while increasing the fitness of others. Ramsey used repertoires instead of concrete acts because he views fitness as a propensity that doesn’t change over the life of an organism (Ramsey 2006). For our purposes here, we need not worry about this subtlety.

What is important is that this notion of altruism is based on biological fitness, which is based on reproduction. It is not linked to the psychological characteristics of the altruists. In fact, creatures lacking anything like intentions in a representationalist sense—plants, for instance—can be biologically altruistic. Nor is it based on the non-reproduction-based positive or negative impacts it has for other individuals, such as helping an individual solve a problem or feel better.

Next consider psychological altruism. This is altruism based on the (mentally formed) intentions of the actor, not the outcomes of their behavior. If I intend to help an octogenarian by offering to assist them in crossing the street, this is not linked to my or their biological fitness. And it may not actually help. The octogenarian may be insulted and not grateful, for instance.

Finally, consider what Ramsey labeled “helping altruism.” This is altruism measured not in terms of intentions or biological fitness, but merely in terms of outcomes. If the octogenarian in the above example is helped across the street—and if this was genuinely helpful—this is an altruistic act even if it has nothing to do with fitness, and even if I did it for selfish reasons, not because I actually wanted to help the person out.

These three forms of altruism can be inverted to obtain three forms of selfishness. First, there is biological selfishness. Borrowing from the definition of biological altruism, an act is *biologically selfish* if and only if having the act in the behavioral repertoire of the individual (1) increases its fitness and (2) decreases the fitness of the recipient(s) (or the group to which the selfish agent belongs).

Psychological selfishness would involve performing acts that are intended to benefit the actor at the expense of others. And because the opposite of helping is hindering, we could label the third form of selfishness as *hindering selfishness*.

Now let us consider Hendrycks's (2023) worry that selfishness is something likely to evolve, and that it will tend to be harmful. Biological selfishness is not an outcome but a relationship among selection pressures. It is also not clear that biological selfishness is more dangerous than biological altruism. In honeybees, for instance, workers carry barbed stingers that make their attacks suicidal. If you brush one off after it stings, its stinger stays intact and serves as a beacon for other attacking bees. This altruistic behavior on the part of bees makes them *more* dangerous, not less.

The observation that altruism is dangerous goes back as far as Darwin (1871). He pondered the question of why brave warriors should exist. Isn't it better to be selfish and hold back in battle? He pointed out that the solution to this comes from considering the benefits to the group, not just the individual. A trait could be bad at the individual level but because of the benefits to the group, it could nevertheless spread in the population. This paradoxical situation is an instance of Simpson's paradox.

Hendrycks (2023) argues that from an information point of view, if the information (in this case, the trait) increases, it is selfishness. However, to examine only what happens overall is to commit the *averaging fallacy* (Sober and Wilson 1998). Selfishness and altruism are not about what happens *on average*, but about the selection pressures in group-structured populations. Biological selfishness, therefore, is neither an inevitable outcome of evolution by natural selection, nor a reliable proxy for AI risks.

Next, consider psychological selfishness. Might this be the problematic intermediate? Though in some cases we may feel justified in attributing selfish intentions or other mental states to artificial systems, Hendrycks (2023) is clear that the phenomenon he is concerned with is not psychological. This allows him to discuss the risks of AI without answering the vexing question of how to characterize the psychological states of AI systems. Psychological selfishness is thus not the intermediate he is concerned with.

This leaves hindering selfishness. Hindering is not an automatic outcome of evolution nor is the harmfulness of it an open empirical question. Hindering is, by definition, harmful.

As we established in Section 3, a concept that is definitionally synonymous with harm cannot act as an intermediate between evolution and harm. It is thus not a contender for an intermediate step between evolution and safety.

Selfishness, we thus conclude, is not an intermediate between evolution and AI safety. And as we will argue in the next section, it is not what makes evolutionary AI dangerous. Evolution does bring new risks to AI, but not because of selfishness. Selfishness is therefore merely an unhelpful distraction.

5. Selfishness is not a risk factor, but evolution is

We have argued that selfishness is not the means by which evolution undermines AI safety. Dismissing “selfishness” as the important risk factor does not mean dismissing the dangers of AI evolution. In this section, we will offer an account of why evolutionary AI poses safety risks.

5.1 Evolutionary risk is not an optimization risk

At this point, it is crucial to distinguish evolution from learning, even though both may be described as search processes (Breedon 2025). In most AI contexts, learning refers to within-individual optimization: a model updates its parameters, policy, or even representations through training, feedback, or interaction with an environment (Sutton and Barto 2018).

Evolution, by contrast, is a population-level process. As we saw in section 2, a basic requirement for independent evolution to occur is that the system be able to autonomously replicate. Selection requires a population of systems that vary in their characteristics, and that some of this variation impacts the systems’ ability to survive and reproduce. While this is sufficient for selection, to achieve evolution by natural selection, some of the characteristics of the replicating entities must be inherited. Evolution by natural selection thus requires reproduction, variation, fitness differences, and inheritance.

The concern here is not merely that individual AI systems may learn (thereby acquiring variations) or become more capable, but that populations of AI systems capable of replication, inheritance, and variation may undergo selection. What matters is not just optimization within a single AI agent, but open-ended population-level adaptation.

This distinction can be clarified by contrasting two familiar AI-risk narratives: the paperclip maximizer and the gray-goo scenario. The paperclip maximizer is a story about the dangers of unconstrained optimization (Bostrom 2020). The system remains perfectly faithful to a fixed, human-specified objective—maximizing the number of paperclips in

the universe—yet, as its capabilities grow, it acquires increasingly effective instrumental strategies for achieving that end. It may seek resources, preserve itself, manipulate humans, or expand its control, not because this is an outcome of evolution, but because such strategies are useful means of better optimizing its assigned objective. The central danger is therefore not evolutionary but rather the catastrophic consequences of extreme optimization under a badly specified goal.

The gray-goo scenario is importantly different. Its threat does not arise from the relentless optimization of a single externally imposed objective, but from the dynamics of self-replication, population growth, and material conversion in an open environment (Drexler 1986). Once such systems can build further copies of themselves from ambient resources, the relevant danger is not simply that they “want” something harmful or act “selfishly,” but that they can expand as a reproducing lineage: their numbers increase, their material footprint grows, and their continued persistence depends on redirecting more of the surrounding world into forms usable for further replication. In that sense, gray goo is threatening not as a single misaligned optimizer, but as an emerging machine population whose proliferation can displace, consume, and ultimately overwhelm existing biological life.

This contrast sharpens the point at issue. Not all catastrophic AI risk is evolutionary risk. Some risks arise from the optimization of a fixed objective by a single highly capable system. Evolutionary risk, by contrast, emerges when populations of systems are able to replicate, vary, inherit traits, and compete.

5.2 Niche prospecting as the true danger of evolutionary AIs

Once evolutionary risk is distinguished from ordinary optimization risk, the next question is what makes it dangerous in its own right. We should thus ask if any of the conditions for evolution by natural selection are dangerous on their own, as well as what dangers emerge from their combination. For instance, the replication of autonomous systems has already emerged as a major safety focus, since it is increasingly treated as a central dimension of model autonomy and loss-of-control risk (OpenAI 2025; AISI 2025). Nevertheless, the real danger of evolutionary AI systems comes from replication in conjunction with the other factors linked to natural selection. To see how the systems become dangerous, consider what evolution by natural selection allows populations of organisms to accomplish.

All organisms occupy a niche. Wildebeest live in groups and graze in savannas. Hawk moths hover like hummingbirds and drink nectar from flowers. Evolution by natural selection allows such species to maintain their niche (via stabilizing selection) or to change

niches (via directional selection). Take the Triatominae, a subfamily of assassin bugs (Reduviidae), which mostly feed on vertebrates (Monteiro et al. 2018). Their ancestors fed on invertebrates but they, at some point, found a new rich store of nutrients, vertebrate blood. Some of these tiny “vampires,” such as the species *Rhodnius prolixus*, carry Chagas disease, which has infected millions of humans (Mesquita et al. 2015). They are sometimes referred to as “kissing bugs”—the name deriving from their proclivity to feed from or near human lips.

This transformation of their niche—switching from feeding on invertebrates to feeding on vertebrates—was not good for humans. Whether or not we should consider these bugs to be selfish is beside the point. What is important is that they moved into a niche that is harmful to us.

While natural selection is often invoked to account for adaptations, for the fit between organism and environment, the inherent production of variation in the process of evolution by natural selection leads species to explore the space of possible niches. Within the Hemiptera (true bugs), there are those that use their needle mouths to feed on the phloem of plants, others that feed on other insects (like many of the Reduviidae just discussed), and the ones that drink vertebrate blood. The proliferation of species has allowed for the exploration of the space of possible nutrient-rich liquids. Some species have no impact on humans, some are beneficial to us (e.g., some *Orius* species—“minute pirate bugs”—are raised to eat agricultural pests, such as plant lice), others are plant pests, still others can transmit devastating diseases.

We contend that this exploration of niche space, which we will call *niche prospecting*, is the true danger of evolutionary AI. By a niche, we simply mean a relational arrangement through which a system persists and reproduces, including both the conditions it exploits and those it actively reshapes (Lewontin 1983; Odling-Smee et al. 2003; Aaby and Ramsey 2022; Ramsey and Fábregas-Tejeda, 2026). In AI systems, niches need not be biological. They are, generally speaking, “job descriptions” for organisms, AI agents, or other systems. They may consist in possible ways of securing, not the nutrient-rich liquids of the Hemiptera, but other “nutrient-rich resources,” such as compute, electricity, API access, financial resources, deployment opportunities, user dependence, tool access, or institutional protection.²

As AI populations undergo evolution by natural selection, they will prospect for new

². This way of speaking is compatible with formal work on power-seeking, where agents tend to favor states that preserve or expand future possibilities (Turner et al. 2021), though the present notion of niche is broader and concerns evolving populations and multi-agent systems instead of individual policies in toy settings.

niches. Like with the Hemiptera, some will be beneficial to us, some neutral, others dangerous. The more these systems are able to prospect, especially prospecting in very different kinds of niches, the more probable it will be for some of the systems to be harmful to humans. Niche prospecting, then, is the true danger of evolutionary AI.

6. Conclusions

While some AI researchers see the production of *selfishness* as the reason evolutionary AI systems are dangerous, we have argued that this is not the case. There are three key meanings of selfishness: biological selfishness, psychological selfishness, and hindering selfishness. None of these is the appropriate causal intermediate between evolution and AI safety. Instead, we identified the key danger of evolutionary AI as *niche prospecting*, which involves exploring the space of possible niches. While many of these niches will be beneficial or neutral to humans, many will be harmful. If there are more ways for an AI system to be harmful to humans than helpful—as seems a reasonable assumption—then niche prospecting will tend to be bad for humans. Because evolution by natural selection involves niche prospecting, allowing AI systems to autonomously evolve by natural selection will tend to pose risks for AI safety.

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References

- Aaby, B.H. and Ramsey, G., 2022. Three kinds of niche construction. *The British Journal for the Philosophy of Science*. <https://doi.org/10.1093/bjps/axz054>.
- AI Security Institute. (2025). *Frontier AI Trends Report*. (Published December 18, 2025). <https://www.aisi.gov.uk/frontier-ai-trends-report>.
- Bereska, L., & Gavves, E. (2024). Mechanistic interpretability for AI safety--a review. *arXiv preprint arXiv:2404.14082*. <https://doi.org/10.48550/arXiv.2404.14082>.
- Black, S., Stickland, A. C., Pencharz, J., Sourbut, O., Schmatz, M., Bailey, J., ... & Cooney, A. (2025). Replibench: Evaluating the autonomous replication capabilities of language model agents. *arXiv preprint arXiv:2504.18565*. <https://doi.org/10.48550/arXiv.2504.18565>.

- Bostrom, N. (2020). "The Ethical Issues of Advanced Artificial Intelligence." In W. Wallach & P. Asaro (Eds.), *Machine Ethics and Robot Ethics* (pp. 69-75). Routledge. (Original work published 2003)
- Breeden, J. L. (2025). The evolution of goals in AI agents. *AI and Ethics*, 5(4), 3897-3915. <https://doi.org/10.1007/s43681-025-00691-y>.
- Chan, A., Salganik, R., Markelius, A., Pang, C., Rajkumar, N., Krasheninnikov, D., ... & Krueger, D. (2023). Harms from increasingly agentic algorithmic systems. In *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency* (pp. 651-666).
- Darwin, C. (1871). *The descent of man, and Selection in relation to sex* (Vols.1-2). John Murray.
- Dawkins, R. (1983). Universal Darwinism. In D.S. Bendall (Ed.), *Evolution from Molecules to Men* (pp.403-425). Cambridge University Press.
- Dolson, E., & Ofria, C. (2021). Digital evolution for ecology research: a review. *Frontiers in Ecology and Evolution*, 9, 750779. <https://doi.org/10.3389/fevo.2021.750779>.
- Doshi-Velez, F., & Kim, B. (2017). Towards a rigorous science of interpretable machine learning. arXiv preprint arXiv:1702.08608. <https://doi.org/10.48550/arXiv.1702.08608>.
- Drexler, E. (1987). *Engines of creation: The coming era of nanotechnology*. Anchor Press.
- Fernando, C., Banarse, D., Michalewski, H., Osindero, S., & Rocktäschel, T. (2023). Promptbreeder: Self-referential self-improvement via prompt evolution. arXiv preprint arXiv:2309.16797. <https://doi.org/10.48550/arXiv.2309.16797>.
- Gao, H. A., Geng, J., Hua, W., Hu, M., Juan, X., Liu, H., ... & Wang, M. (2026) A Survey of Self-Evolving Agents: What, When, How, and Where to Evolve on the Path to Artificial Super Intelligence. *Transactions on Machine Learning Research*. <https://doi.org/10.48550/arXiv.2507.21046>.
- Godfrey-Smith, P. (2007). Conditions for evolution by natural selection. *The Journal of Philosophy*, 104(10), 489–516. <https://doi.org/10.5840/jphil2007104103>
- Hammond, L., Chan, A., Clifton, J., Hoelscher-Obermaier, J., Khan, A., McLean, E., ... & Rahwan, I. (2025). Multi-agent risks from advanced ai. arXiv preprint arXiv:2502.14143. <https://doi.org/10.48550/arXiv.2502.14143>.
- Hendrycks, D. (2023). Natural selection favors AIs over humans. arXiv preprint arXiv:2303.16200. <https://doi.org/10.48550/arXiv.2303.16200>.

- Hendrycks, D., Mazeika, M., & Woodside, T. (2023). An overview of catastrophic AI risks. arXiv preprint arXiv:2306.12001. <https://doi.org/10.48550/arXiv.2306.12001>.
- Kinniment, M., Chan, J. S., Radtke, T., MacDiarmid, M., Lin, J., Bai, A., ... & Barnes, L. (2023). Evaluating language-model agents on realistic autonomous tasks. arXiv preprint arXiv:2312.11671. <https://arxiv.org/abs/2312.11671>.
- Lehman, J., Gordon, J., Jain, S., Ndousse, K., Yeh, C., & Stanley, K. O. (2022). Evolution through large models. arXiv preprint arXiv:2206.08896. <https://arxiv.org/abs/2206.08896>.
- Lewontin, R. C. (1970). The units of selection. *Annual review of ecology and systematics*, 1, 1-18. <https://doi.org/10.1146/annurev.es.01.110170.000245>.
- Lewontin, R. C. (1983). The organism as the subject and object of evolution. *Scientia*, 118, 63-82.
- Linardatos, P., Papastefanopoulos, V., & Kotsiantis, S. (2020). Explainable ai: A review of machine learning interpretability methods. *Entropy*, 23(1), 18. <https://doi.org/10.3390/e23010018>.
- Mesquita, R. D., Vionette-Amaral, R. J., Lowenberger, C., Rivera-Pomar, R., Monteiro, F. A., Minx, P., ... & Oliveira, P. L. (2015). Genome of *Rhodnius prolixus*, an insect vector of Chagas disease, reveals unique adaptations to hematophagy and parasite infection. *Proceedings of the National Academy of Sciences*, 112(48), 14936-14941. <https://doi.org/10.1073/pnas.1506226112>.
- Monteiro, F. A., Weirauch, C., Felix, M., Lazoski, C., & Abad-Franch, F. (2018). Evolution, systematics, and biogeography of the Triatominae, Vectors of Chagas disease. *Advances in parasitology*, 99, 265-344. <https://doi.org/10.1016/bs.apar.2017.12.002>.
- Moritz, P., Nishihara, R., Wang, S., Tumanov, A., Liaw, R., Liang, E., Elibol, M., Yang, Z., Paul, W., Jordan, M. I., & Stoica, I. (2018). Ray: A distributed framework for emerging AI applications. *Proceedings of the 13th USENIX Symposium on Operating Systems Design and Implementation (OSDI 18)*, 561-577.
- Odling-Smee, F. J., Laland, K. N., & Feldman, M. W. (2003). *Niche construction: The neglected process in evolution*. Princeton University Press.
- OpenAI. (2025). Preparedness Framework (Version 2, updated April 15, 2025). <https://openai.com/index/updating-our-preparedness-framework/>.
- Pan, X., Dai, J., Fan, Y., & Yang, M. (2024). Frontier AI systems have surpassed the self-replicating red line. arXiv preprint arXiv:2412.12140. <https://doi.org/10.48550/arXiv.2412.12140>.

- Papale, F. (2021). Evolution by means of natural selection without reproduction: revamping Lewontin's account. *Synthese*, 198(11), pp.10429-10455. <https://doi.org/10.1007/s11229-020-02729-6>.
- Rahwan, I., Cebrian, M., Obradovich, N., Bongard, J., Bonnefon, J. F., Breazeal, C., ... & Wellman, M. (2019). Machine behaviour. *Nature*, 568, 477-486. <https://doi.org/10.1038/s41586-019-1138-y>.
- Ramsey, G. (2006). Block fitness. *Studies in History and Philosophy of Science Part C: Studies in History and Philosophy of Biological and Biomedical Sciences*, 37(3), 484-498. <https://doi.org/10.1016/j.shpsc.2006.06.009>.
- Ramsey, G. (2016). Can altruism be unified?. *Studies in History and Philosophy of Science Part C: Studies in History and Philosophy of Biological and Biomedical Sciences*, 56, 32-38. <https://doi.org/10.1016/j.shpsc.2015.10.007>.
- Ramsey, G. and Fábregas-Tejeda, A. (2026). The Matrix Account of Niche Construction. PhilSci-Archive. <https://philsci-archive.pitt.edu/30043/>.
- Ray, T. S. (1992). *Evolution, ecology and optimization of digital organisms* (Santa Fe Institute Working Paper No. 92-08-042). Santa Fe Institute. <https://www.santafe.edu/research/results/working-papers/evolution-ecology-and-optimization-of-digital-orga>.
- Russell, S. (2019). *Human compatible: AI and the problem of control*. Penguin.
- Shen, Z. (2024). LLM with tools: A survey. arXiv preprint arXiv:2409.18807. <https://doi.org/10.48550/arXiv.2409.18807>.
- Sober, E. and Wilson, D.S., 1998. *Unto Others: The Evolution and Psychology of Unselfish Behavior*. Harvard University Press.
- Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction* (2nd ed.). MIT Press.
- Turner, A. M., Smith, L., Shah, R., Critch, A., & Tadepalli, P. (2021). Optimal policies tend to seek power. *Advances in Neural Information Processing Systems*, 34, 23063–23074.
- Von Neumann, J. (1966). *Theory of self-reproducing automata* (A. W. Burks, Ed.). University of Illinois Press.
- Waldner, D. (2025). AI Safety, Alignment, and Ethics (AI SAE). arXiv preprint arXiv:2509.24065. <https://doi.org/10.48550/arXiv.2509.24065>.

- Wilke, C. O., & Adami, C. (2002). The biology of digital organisms. *Trends in ecology & evolution*, 17(11), 528-532. [https://doi.org/10.1016/S0169-5347\(02\)02612-5](https://doi.org/10.1016/S0169-5347(02)02612-5).
- Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., ... & Rush, A. M. (2019). Huggingface's transformers: State-of-the-art natural language processing. arXiv preprint arXiv:1910.03771. <https://doi.org/10.48550/arXiv.1910.03771>.
- Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., Jiang, L., Zhang, X., Zhang, S., Liu, J., Awadallah, A. H., White, R. W., Burger, D., & Wang, C. (2023). AutoGen: Enabling next-gen LLM applications via multi-agent conversation. arXiv preprint arXiv: 2308.08155. <https://doi.org/10.48550/arXiv.2308.08155>.
- Wulf, J., Meierhofer, J., & Hannich, F. (2025). Architecting Human-AI Cocreation for Technical Services—Interaction Modes and Contingency Factors. arXiv preprint arXiv: 2507.14034. <https://doi.org/10.48550/arXiv.2507.14034>.
- Zaharia, M., Chen, A., Davidson, A., Ghodsi, A., Hong, S. A., Konwinski, A., Murching, S., Nykodym, T., Ogilvie, P., Parkhe, M., Xie, F., & Zumar, C. (2018). Accelerating the machine learning lifecycle with MLflow. *IEEE Data Engineering Bulletin*, 41(4), 39–45. <https://doi.org/10.1145/3399579.3399867>.